TOBE ENERGY

Techno-Economic Analysis

Repeat-Commercial Distributed Hydrogen Facility

~5 MW-class | 900,000 kg/yr nameplate | 200 bar delivered

Confidentiality and estimate basis

Prepared for investor evaluation and strategic diligence. The anchor case excludes Section 45V value, excludes terminal value, and treats the Zeeco FOAK project as an engineering validation case rather than the normalized commercial benchmark. Capital is based on the current live internal AFE and should be regarded as budgetary-level with expected accuracy on the order of ±20–30% until refreshed by a formal third-party estimate.

March 2026

Executive Summary

This TEA evaluates a repeat-commercial distributed hydrogen facility built around Tobe Energy's isothermal electrolysis platform. The governing commercial case is a fully built facility producing 900,000 kg/yr nameplate and 810,000 kg/yr at steady state, delivering hydrogen at ≥99.95% purity (dry basis) and 200 bar within a 50 km radius by tube trailer. The purpose is to determine whether that repeat-commercial configuration supports attractive economics on an unsubsidized basis.

On the Oklahoma reference power basis used in the current financial model, the answer is yes. Direct delivered variable cost is approximately $2.57/kg, cash operating cost is approximately $4.62/kg, and all-in economic cost including annualized capital is approximately $5.25/kg. At a $25/kg delivered selling price, steady-state site EBITDA is approximately $16.5 million per year. The downside case—$20/kg selling price, 85% availability, and maintenance at the upper consultant reference range—remains clearly EBITDA positive.

The main diligence question is therefore not whether the economics depend on policy support. They do not. The main question is whether Tobe can repeat the modeled performance, capital intensity, and price capture as it moves from the Zeeco FOAK platform to repeat-commercial deployments built around larger modules and standardized site packages.

Anchor-case metric	Value	Interpretation
Commercial product	Delivered H_2, ≥99.95% dry, 200 bar, ≤50 km	Governing repeat-commercial boundary
Steady-state output	810,000 kg/yr	900,000 kg/yr nameplate at 90% availability
Direct delivered variable cost	$2.57/kg	Electricity, compression, logistics, and water
Cash operating cost	**$4.62/kg**	**Direct variable + maintenance + fixed site cost**
All-in economic cost	**$5.25/kg**	**Cash operating cost + annualized capital**
Steady-state revenue	$20.25M/yr	$25/kg delivered selling price
Steady-state EBITDA	**$16.51M/yr**	**81.5% site EBITDA margin**
Downside case	**$11.58M/yr EBITDA**	**$20/kg, 85% availability, $0.57/kg maintenance**
Capital basis	**$4.575M fully built equivalent**	**Live AFE retained as conservative build basis**

This document deliberately separates three concepts that were blurred in earlier drafts: (i) the repeat-commercial delivered-hydrogen case that anchors the economics, (ii) the Zeeco FOAK burner-supply project that anchors engineering credibility, and (iii) optional higher-pressure fueling or mobility configurations that are not used in the anchor economics.

1. Facility Definition and Product Boundary

Boundary discipline is the most important framing choice in this TEA. The attached source set describes several valid project configurations, but they are not interchangeable. The commercial case used here is a delivered-hydrogen facility. The Zeeco project is a separate FOAK integration case. Optional fueling and dispensing configurations are future commercial options and are excluded from the anchor economics.

Case	Use in this document	Product specification	Delivery / offtake boundary
Repeat-commercial anchor case	Main body and return metrics	≥99.95% dry H_2 at 200 bar; tube-trailer delivered	Delivered within 50 km to undersupplied local demand
Zeeco FOAK validation case	Engineering validation only	Low-pressure direct-to-burner supply in Phase 1	On-site partner integration; no Phase 1 storage
Optional mobility / fueling case	Excluded from anchor economics	350–700 bar, use-case specific purity and dispensing	Future fueling / dispensing configuration only

Included and excluded scope

Included in the anchor case	Excluded from the anchor case
Electrolysis modules and electrical balance of plant Water polishing, purification, drying, and compression to 200 bar Tube-trailer loading and logistics within 50 km Site labor, lease, and current full-build capital basis	Fueling / dispensing beyond the 200 bar delivery boundary Upstream power-generation assets Land acquisition (lease assumed) Corporate overhead, equipment sales, and service revenue Section 45V value and terminal value

The current facility AFE includes some higher-pressure compression, storage, and dispensing line items that are broader than a stripped 200 bar delivered-hydrogen boundary. Rather than remove those items opportunistically without a refreshed EPC estimate, this TEA retains the live AFE as the governing capital basis and treats the resulting $4.575 million as a conservative commercial build benchmark.

2. Technology and Operating Basis

Tobe's platform is presented in company technical materials as a membrane-free, PGM-free, near-ambient electrolysis system built primarily from 304 stainless steel and commodity industrial components. The technical case is attractive because it combines low specific energy consumption, simple materials, low-temperature operation, and modular serviceability. The investment question is not whether those attributes are directionally valuable; it is whether the commercial operating basis used here is conservative, explicit, and consistent.

Operating basis	Value used in this TEA	Comment
Equivalent continuous AC input	~4.4 MW	900,000 kg/yr × 42.8 kWh/kg ÷ 8,760 hours
Canonical system energy basis	42.8 kWh/kg AC-to-product	Commercial governing basis used for cost calculations
Compression increment	3.5 kWh/kg	Compression to the 200 bar delivery boundary
Water consumption	9 L/kg H_2	RO/DI water to product basis
Steady-state availability	90%	Anchor case
Project life	9 years	No terminal value in anchor case
Year 1 ramp	75% of steady-state output	607,500 kg in Year 1, then steady state thereafter
Tax rate	27%	After-tax unlevered return basis

Validation status and treatment in this TEA

Topic	Basis used here	Current status / treatment
T-25 architecture and FOAK battery limits	Current engineering package	Sufficiently evidenced for current TEA purpose

Topic	Basis used here	Current status / treatment
Commercial system energy basis	42.8 kWh/kg AC-to-product	Conservative modeled basis; slightly above the ~42–42.2 kWh/kg values shown in company materials and affiliate quote materials
Stack life treatment	80,000+ hour design-life basis	Modeled commercial basis; long-duration validation should be added to the data room
Field serviceability	30-minute T-25 swap demonstrated	Do not extend the demonstrated swap-time claim to T-125 without supplemental evidence
T-125 repeat-commercial architecture	Scale bridge from current source set	Commercial basis is reasonable but still requires additional validation packaging

For the 9-year anchor case, the current stack-life basis implies no modeled stack replacement reserve. At 90% availability, the facility accumulates approximately 70,956 operating hours over the project life; at 95% availability, approximately 74,898 hours. That treatment is analytically coherent on the current design-life basis, but it remains a modeled commercial assumption rather than a third-party-certified field result.

3. Methodology and Source Hierarchy

This document follows a deliberate evidence hierarchy so that the economics are governed by the strongest current sources rather than by whichever document states the lowest cost or highest performance. The priority is: current financial model for numerical basis; current engineering package for scope and battery limits; current published technical materials for technology description and standards; affiliate and consultant quote materials only as supplementary context or sensitivity support.

Source class	Primary role in the TEA	Treatment
Current financial model	Numerical basis for production, pricing, labor, lease, and live AFE	Controls anchor-case numbers unless the TEA explicitly states a more conservative governing basis
Current engineering package	FOAK scope split, battery limits, electrical basis, and process architecture	Controls Zeeco boundary definition and validation framing
Published Tobe technical materials	Technology description, controls, utility requirements, codes, and deployment logic	Used for narrative and technical context
Affiliate / consultant quote materials	Supplementary equipment and maintenance references	Used only as corroboration, sensitivity input, or directional context; not as the sole basis for normalized economics
Evidenced today	**Modeled commercial basis**	**Intentionally excluded from the anchor case**
FOAK battery limits and scope split T-25 prototype operating architecture Current personnel and lease basis Live AFE as current capital source	42.8 kWh/kg governing energy basis 80,000+ hour stack-life basis T-125 repeat-commercial scale bridge No stack replacement inside 9-year base life	Section 45V value Terminal value Corporate overhead Equipment sales and service revenue Optional fueling / dispensing economics

Return metrics are calculated on an after-tax unlevered basis using the fully built $4.575 million capital base, 27% tax, 9-year life, and 75% of steady-state output in Year 1. No terminal value is used. The base case excludes all Section 45V value by design.

4. Cost Build-Up

4.1 Direct delivered variable cost

Component	Formula basis	$/kg	Annual at 810,000 kg/yr
Electrolysis	42.8 kWh/kg × $0.035/kWh	$1.50	$1.213M
Compression	3.5 kWh/kg × $0.035/kWh	$0.12	$0.099M
Logistics	Tube trailer delivery within 50 km	$0.90	$0.729M
Water	9 L/kg × $0.00556/L	$0.05	$0.041M
Direct delivered variable cost	**Sum of the above**	**$2.57**	**$2.082M**

4.2 Fixed operating cost and capital basis

Item	Basis	Annual or total value
Facility operations crew	Current loaded seven-person site crew basis	$1.214M / year
Site lease (Tulsa reference)	Current model basis	$0.105M / year
Fixed site operating cost	**Labor + lease**	**$1.319M / year**
Current additional facility AFE	Live AFE base tranche	$3.075M
Expansion tranche	Current additional facility upgrade cost	$1.500M
Fully built capital basis	**Base tranche + expansion tranche**	**$4.575M**

4.3 Anchor-case cost stack

Cost layer	$/kg
Direct delivered variable cost	$2.57
Maintenance (anchor-case model basis)	$0.42
Fixed site operating cost	$1.63
Cash operating cost	**$4.62**
Annualized capital	$0.63
All-in economic cost	**$5.25**

The maintenance anchor uses the current financial-model basis of $0.42/kg. A higher consultant-prepared range of $0.52–$0.57/kg exists in the source materials and is used here only as a reference point and downside sensitivity, not as the governing normalized base case.

5. Facility Economics

5.1 Steady-state operating profile

Line item	Annual value	$/kg
Revenue	$20.250M	$25.00
Direct delivered variable cost	($2.082M)	($2.57)
Maintenance	($0.340M)	($0.42)

Line item	Annual value	$/kg
Fixed site operating cost	($1.319M)	($1.63)
Site EBITDA	**$16.509M**	**$20.38**
EBITDA margin	**81.5%**	**—**

5.2 Condensed after-tax cash flow

Period	Output (kg)	Revenue	EBITDA	After-tax operating cash flow
Year 0	—	—	—	($4.575M) capital outlay
Year 1	607,500	$15.188M	$12.052M	$8.935M
Years 2–9 (each)	810,000	$20.250M	$16.509M	$12.189M

Return metrics are intentionally shown only after the operating basis is established

On this after-tax unlevered basis, the anchor case generates an IRR of approximately 217.7% and an NPV of approximately $57.5 million at a 12% discount rate. These outputs follow mechanically from a small capital base and a high delivered-price market. They should be read as a consequence of the assumed business model, not as a substitute for validating repeatability.

A more useful summary statistic than the headline IRR is the hurdle selling price. Under the anchor-case cost structure, a delivered price of roughly $6/kg is sufficient to clear a 12% after-tax unlevered hurdle. That is materially below the $25/kg anchor case and still below the downside price case.

6. Downside Resilience and Market Sensitivity

6.1 Scenario analysis

Scenario	Price	Availability	Steady output	Maintenance	Cash cost	All-in cost	EBITDA
Downside	$20/kg	85%	765,000	$0.57/kg	$4.80/kg	$5.46/kg	$11.58M
Base	$25/kg	90%	810,000	$0.42/kg	$4.62/kg	$5.25/kg	$16.51M
Upside	$30/kg	95%	855,000	$0.40/kg	$4.51/kg	$5.11/kg	$21.81M

The downside case is the most important line in the table. It assumes lower price, lower availability, and maintenance at the upper consultant reference range. Even in that case the site remains strongly cash-generative. That is why this should be treated as a distributed local-market infrastructure thesis rather than as a tax-credit life-support story.

6.2 Electricity-price sensitivity across target markets

Market	Power rate	Direct delivered variable cost	Cash operating cost	All-in economic cost
Tulsa, OK	$0.0350	$2.57/kg	$4.62/kg	$5.25/kg
Oklahoma City, OK	$0.0350	$2.57/kg	$4.62/kg	$5.25/kg
Spokane, WA	$0.0413	$2.86/kg	$4.95/kg	$5.58/kg
Des Moines, IA	$0.0429	$2.94/kg	$4.99/kg	$5.61/kg
Albuquerque, NM	$0.0587	$3.67/kg	$5.74/kg	$6.37/kg
Seattle, WA	$0.0595	$3.70/kg	$5.81/kg	$6.44/kg

Power price matters, but it does not overturn the thesis under the current delivered-price assumption. The spread between Oklahoma and the higher-cost target markets is meaningful, yet the modeled economics remain robust because the delivered-price thesis is driven by local merchant scarcity rather than by national commodity pricing.

6.3 Pricing interpretation

The $25/kg anchor price is not a claim about the national commodity hydrogen market. It is a site-level delivered-price assumption for structurally undersupplied local demand pockets that often rely on small-volume merchant supply, specialty delivery, or cylinder replacement rather than pipeline-scale hydrogen. The commercial support behind that assumption should be packaged separately in the data room through an anonymized pricing appendix showing geography, delivered specification, volume context, quote type, and commercial status. This TEA therefore treats $25/kg as the canonical commercial case and separately tests resilience at $20/kg.

7. Lifecycle, Durability, and Scale Path

7.1 Lifecycle implication of the current stack-life basis

Metric	Value	Implication
9-year operating hours at 90% availability	70,956 hours	Within the current 80,000+ hour design-life basis
9-year operating hours at 95% availability	74,898 hours	Still within the current design-life basis
Modeled stack replacement in anchor case	None	No replacement reserve is included inside the 9-year base life

7.2 T-25 to T-125 scale bridge

Dimension	T-25 evidenced today	T-125 repeat-commercial basis	Treatment in this TEA
Electrochemical architecture	Current FOAK basis	Same chemistry; larger commercial stack	Reasonable design-basis bridge
Power electronics	48 kW package level	Scaled commercial power stage	Commercial basis, not yet separately field packaged
Serviceability	30-minute module swap demonstrated	Swap timing not yet documented	Do not assume parity without supplemental evidence
Validation status	Prototype/FOAK evidence available	Commercial validation package still needed	Key diligence packaging item

7.3 Scale economics

Metric	Half-cap initial module	Full-cap repeat facility	Improvement at full build
Capex intensity ($ per annual kg)	$6.83	$5.08	−25.6%
Fixed operating burden ($/kg)	$3.26	$1.63	−50.0%
Annualized capital burden ($/kg)	$0.84	$0.63	−25.6%
Cash operating cost ($/kg)	$6.25	$4.62	−26.1%
All-in economic cost ($/kg)	$7.09	$5.25	−26.0%

This is the core modular thesis in economic form: the repeat-commercial facility is not simply a larger copy of a pilot. It is a meaningfully better unit-economic configuration because capital intensity and fixed-cost absorption both improve as the site is built to its full commercial boundary.

8. Zeeco FOAK Validation Case

The Zeeco project should strengthen the investment case by showing engineering integration discipline, not by serving as the normalized economic benchmark. Its value lies in defined battery limits, partner-site integration,

utility sizing, and scope clarity. Its product boundary and commercial structure are different from the repeat-commercial delivered-hydrogen case.

FOAK attribute	Current basis
Location	Zeeco Advanced Research Complex, Broken Arrow, Oklahoma
Configuration	12 × T-25 electrolyzers
Capacity	300 kg/day
Electrical basis	576 kW electrolyzer demand + approximately 50 kW auxiliary and building loads
Transformer	750 kVA
Product boundary	Direct to Zeeco burners / flare; low-pressure supply; no Phase 1 storage
Tobe scope	Electrolyzer skids, transformer, 5 kV distribution, power electronics, PLC/HMI, OT, and commissioning
Zeeco scope	Utility service, RO/DI, building/HVAC, purification/dehydration, flare, vent stack, civil, and permitting

For that reason, Zeeco belongs in the TEA as execution and integration evidence only. It should not be used as the normalized pricing anchor, return case, or product-definition benchmark for repeat-commercial economics.

9. Policy Treatment

Section 45V is upside only

The anchor case excludes all Section 45V value. Any policy upside would require a facility-specific carbon-intensity pathway, compliant power procurement, prevailing wage and apprenticeship compliance, and unrelated-party verification. This TEA is intended to stand without that support.

10. Diligence Priorities and Limitations

The economics shown here are sufficient for a serious first-pass diligence discussion, but several additional items would strengthen broad external circulation. These are packaging and validation priorities, not a rejection of the current thesis.

Recommended data-room addition	Why it matters	Effect on the anchor case if not yet packaged
T-125 performance and duration package	The repeat-commercial case depends on the larger-module bridge	Does not change the current TEA basis, but leaves scale-up questions more open than necessary
Third-party performance validation package	Would reinforce efficiency and durability claims	Anchor case remains intact, but technical diligence will focus heavily here
FEED closeout memo or reissued closed-item register	Current FOAK package is still marked issued-for-review	Avoids preventable diligence friction on engineering maturity
Anonymized pricing appendix	Supports the $25/kg local-market thesis with disciplined commercial evidence	Most important commercial packaging item before broad circulation
Maintenance reserve build-up	Would show bottom-up support beneath the $0.42/kg model basis	The TEA already tests higher maintenance in downside sensitivity

Recommended data-room addition	Why it matters	Effect on the anchor case if not yet packaged
Boundary-clean capital memo	Separates 200 bar delivered scope from optional fueling-oriented lines still visible in the live AFE	Would improve precision without weakening the current conservative capital basis

This TEA does not claim that Tobe has already completed third-party bankability work, contracted a nine-year offtake book, or refreshed the repeat-commercial estimate into a stripped final EPC budget. It does claim that, on the current operating and capital basis, the repeat-commercial economics appear compelling without policy support and that the next important step is disciplined packaging of the remaining validation set.

Appendix A. Source Documents Used in This TEA

Document	Role in this TEA	Treatment
Current TEA draft	Structural starting point	Rewritten and re-ordered; not treated as final authority
Notes and revision notes	Commercial framing, boundary corrections, and diligence priorities	Used to improve logic and investor readability
Current financial model	Numerical basis for costs, production, labor, lease, and live AFE	Primary numerical source
Current FEED / PFD / EL / PID package	FOAK scope, process, electrical basis, and battery limits	Primary engineering source for Zeeco validation case
General arrangement drawings	Conceptual layout context	Explicitly not treated as cost or IFC basis
Technology overview and developer guide	Technology description, utility basis, controls, standards, and deployment logic	Used for technical narrative and operating context
Comparative economics materials	Context for relative positioning	Not used as the anchor-case cost basis
Affiliate / consultant quote materials	Supplementary cost and maintenance references	Sensitivity and corroboration only

Appendix B. Base-Case Return Metrics and Hurdle Price

Metric	Value
After-tax unlevered IRR	**217.7%**
NPV at 12% discount rate	**$57.5M**
Hurdle price for 8% after-tax IRR	**$5.73/kg**
Hurdle price for 10% after-tax IRR	**$5.84/kg**
Hurdle price for 12% after-tax IRR	**$5.96/kg**
Hurdle price for 15% after-tax IRR	**$6.15/kg**

Methodology: fully built $4.575 million capital outlay at Year 0; 27% tax; 75% of steady-state output in Year 1; steady-state output thereafter; straight-line depreciation over 9 years; no terminal value; no Section 45V value.

Appendix C. Full Scenario Reference

Scenario	Price	Availability	Steady output	Maintenance	Cash cost	All-in cost	EBITDA	After-tax IRR	NPV @ 12%
Downside	$20/kg	85%	765,000	$0.57/kg	$4.80/kg	$5.46/kg	$11.58M	156.3%	$39.1M
Base	$25/kg	90%	810,000	$0.42/kg	$4.62/kg	$5.25/kg	$16.51M	217.7%	$57.5M
Upside	$30/kg	95%	855,000	$0.40/kg	$4.51/kg	$5.11/kg	$21.81M	282.6%	$77.2M